

Independent and dependent variables

Bivariate data has two variables that are often identified as the independent and dependent variables. The independent variable is the input. It is not affected by the other variable and is represented on the horizontal axis of the scatterplot. The dependent variable is the output and is 'dependent' on the independent variable. It is represented on the vertical axis of a scatterplot.



Example 3: Identifying independent and dependent variables

6B

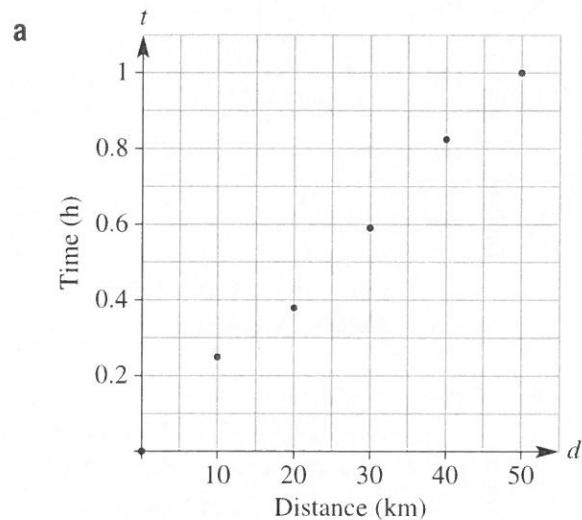
The table below shows the time taken (hours) relative to the distance travelled (km).

Distance (d)	0	10	20	30	40	50
Time (t)	0	0.25	0.38	0.59	0.82	1.00

- Draw a scatterplot using the above table.
- Which are the independent and dependent variables?

SOLUTION:

- Draw a number plane with d as the horizontal axis and t as the vertical axis.
- Determine a scale for the horizontal axis. Let each unit represent 10km.
- Determine a scale for the vertical axis. Let each unit represent 0.2 hours.
- Write titles for the horizontal and vertical axes.
- Plot the points (0, 0) (10, 0.25) (20, 0.38) (30, 0.59) (40, 0.82) (50, 1).



- The independent variable is the input and represented on the horizontal axis of the scatterplot.
- The dependent variable is the output and represented on the vertical axis of the scatterplot.

- b** Independent variable is distance (d).

Dependent variable is time (t).

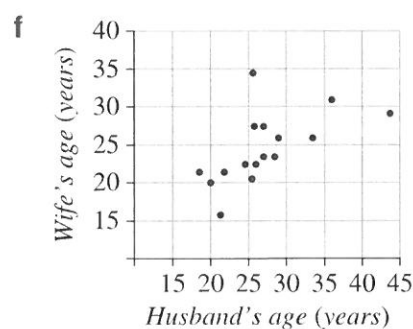
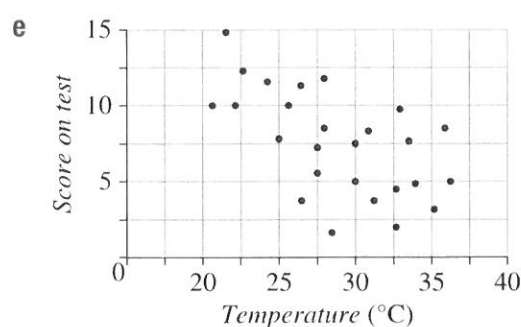
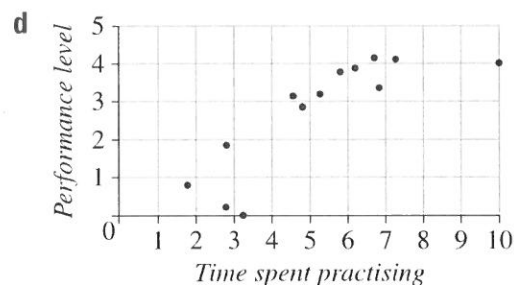
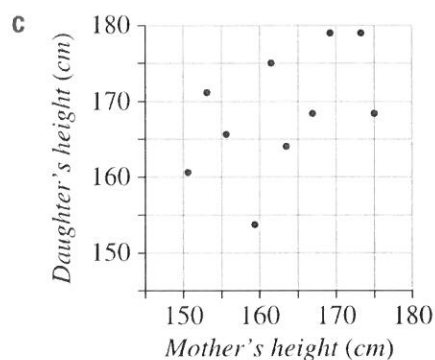
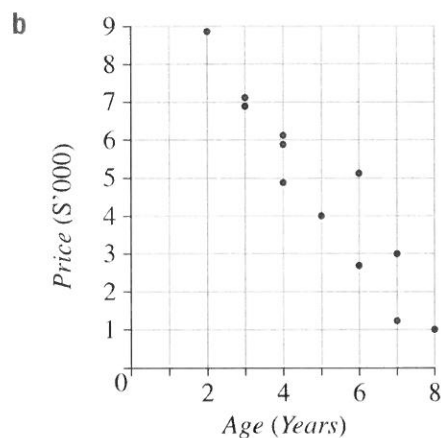
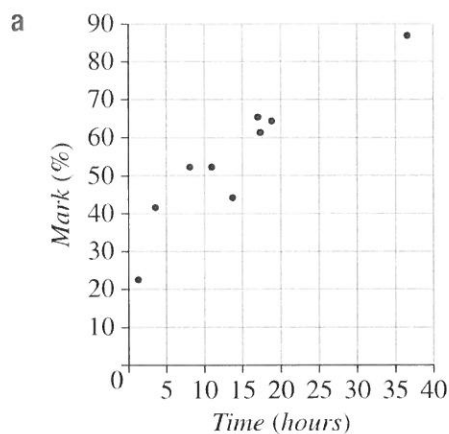
Exercise 6B

LEVEL 1

Example 2

1 Describe the association in the following scatterplots as:

- i linear or non-linear
- ii positive or negative
- iii strong, moderate or weak.



2 For each of the following pairs of variables, indicate whether you expect an association to exist and, if so, whether you would expect the association to be positive or negative.

- | | |
|---|-----------------------------------|
| a Independent variable: Distance travelled | Dependent variable: Time taken |
| b Independent variable: Amount of daily exercise | Dependent variable: Fitness level |
| c Independent variable: Foot length of an adult | Dependent variable: Intelligence |
| d Independent variable: Number of pages | Dependent variable: Book price |
| e Independent variable: Temperature above 30°C | Dependent variable: Comfort level |

LEVEL 2

- Example 3** 3 Chocolates are sold for \$12 per kg. The table below shows weight against cost.

Weight (w)	1	2	3	4	5
Cost (c)	12	24	36	48	60

- a Which is the independent variable? b Which is the dependent variable?
 c Draw a scatterplot of weight against cost. d Is the form linear or non-linear?
 e Is the direction positive or negative? f Is the strength strong, moderate or weak?
- 4 The table below shows the drug dosage against reaction time.

Drug dosage (d)	0.5	1.0	1.5	2.0	2.5	3.0	3.5	4.0	4.5	5.0	6.0
Reaction time (t)	66	48	35	19	18	17	11	15	10	10	11

- a Which is the independent variable? b Which is the dependent variable?
 c Draw a scatterplot of time against cost. d Is the form linear or non-linear?
 e Is the direction positive or negative? f Is the strength strong, moderate or weak?
- 5 Kayla conducted a science experiment and presented the results in a table.

Mass (m)	3	6	9	12	15
Time (t)	8.2	6.7	5.2	3.7	2.2

- a Which is the independent variable? b Which is the dependent variable?
 c Draw a scatterplot of mass against time. d Is the form linear or non-linear?
 e Is the direction positive or negative? f Is the strength strong, moderate or weak?

LEVEL 3

- 6 The table below shows leg length compared to height.

Leg length (in cm)	83	83	85	87	89	89	92	93	94
Height (in cm)	166	167	170	174	179	178	183	185	188

- a Draw a scatterplot using the above table.
 b Describe the association between the leg length and height.
- 7 The table below shows forearm length and hand length.

Forearm (in cm)	27	28	26	28	29	27	25	26	27
Hand (in cm)	19	18.5	18	19	19.5	18	17.5	18	18.5

- a Draw a scatterplot using the above table.
 b Describe the association between the forearm length and hand length.

